

School of Computer Science and Artificial Intelligence

WEEK-9

Assignment #18

Program : B. Tech (CSE)

Specialization :AIML

Course Title : Competitive Programming

Course Code : 23CS302PC305

Semester : VI

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Batch No. : 33

Assignment 3: Pattern Prefix Matching in Log Analysis (Z-Algorithm)

A system administrator analyzes system logs to detect whether a specific error pattern appears as a prefix at any position in the log file.

To efficiently compare the pattern with all suffixes of the log string, the Z- Algorithm is used.

Task

1. Construct the Z-array for the combined string.
2. Identify positions where the pattern fully matches.
3. Output all matching indices. Operations

- BuildZArray(string)
- FindPattern(text, pattern) Example Test Case 1

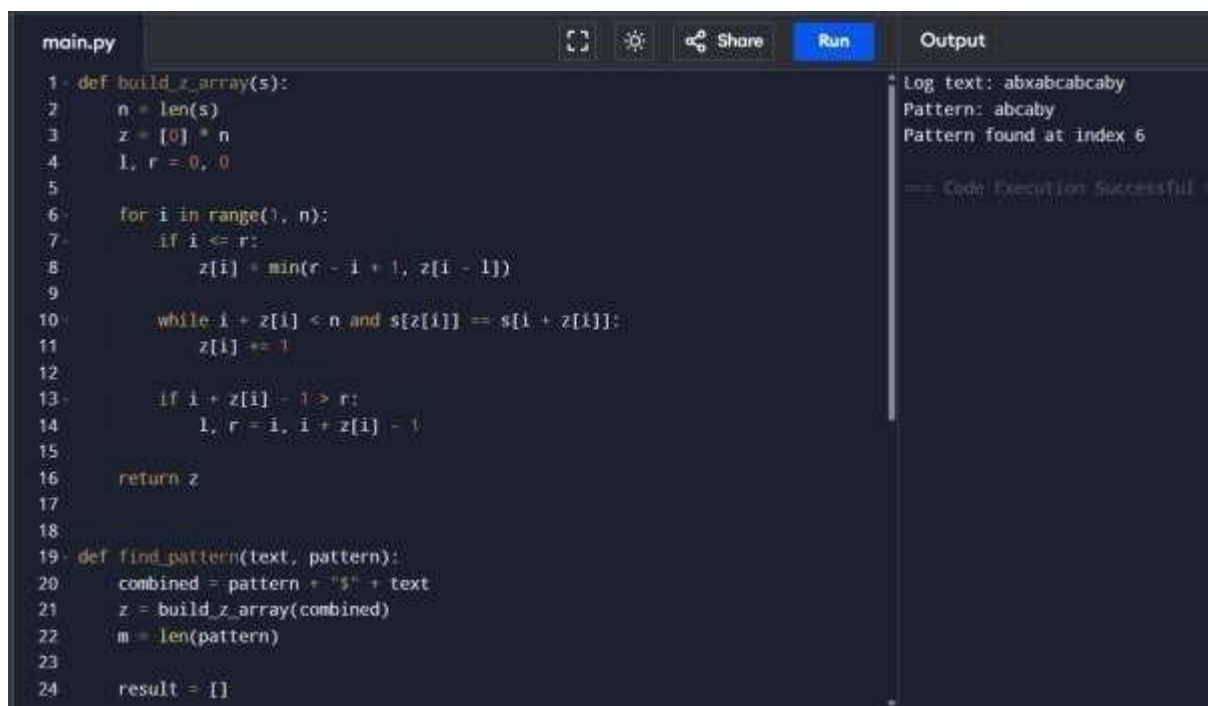
Input

Log text: abxabcabcaby

Pattern: abcaby Output

Pattern found at index 6

PYTHON CODE:



```
main.py  [Icons]  Share  Run  Output

1 def build_z_array(s):
2     n = len(s)
3     z = [0] * n
4     l, r = 0, 0
5
6     for i in range(1, n):
7         if i <= r:
8             z[i] = min(r - i + 1, z[i - 1])
9
10            while i + z[i] < n and s[z[i]] == s[i + z[i]]:
11                z[i] += 1
12
13            if i + z[i] - 1 > r:
14                l, r = i, i + z[i] - 1
15
16        return z
17
18
19 def find_pattern(text, pattern):
20     combined = pattern + "$" + text
21     z = build_z_array(combined)
22     m = len(pattern)
23
24     result = []

Log text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6

== Code Execution Successful ==
```

```
main.py  [Icons] Share Run Output
21 z = build_z_array(combined)
22 m = len(pattern)
23
24 result = []
25
26 for i in range(len(z)):
27     if z[i] == m:
28         index = i - m + 1
29         result.append(index)
30
31 return result
32
33
34 # ---- MAIN ----
35 text = input("Log text: ").strip()
36 pattern = input("Pattern: ").strip()
37
38 matches = find_pattern(text, pattern)
39
40 if matches:
41     for idx in matches:
42         print("Pattern found at index", idx)
43 else:
44     print("No match found")
```

Log text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6
== Code Execution Successful

JAVA CODE:

```
ZAlgorithmSearch.java
1- import java.util.*;
2
3- public class ZAlgorithmSearch {
4
5-     static int[] buildZArray(String s) {
6-         int n = s.length();
7-         int[] z = new int[n];
8-         int l = 0, r = 0;
9
10-        for (int i = 1; i < n; i++) {
11-            if (i <= r) {
12-                z[i] = Math.min(r - i + 1, z[i - l]);
13-            }
14
15-            while (i + z[i] < n && s.charAt(z[i]) == s.charAt(i + z[i])) {
16-                z[i]++;
17-            }
18
19-            if (i + z[i] - 1 > r) {
20-                l = i;
21-                r = i + z[i] - 1;
22-            }
23-        }
24-    }
}
```

Log text: abxcabcabcy
Pattern: abcaby
Pattern found at index 6

=== Code Execution Successful ===

```
ZAlgorithmSearch.java
43 }
44
45 public static void main(String[] args) {
46     Scanner sc = new Scanner(System.in);
47
48     System.out.print("Log Text: ");
49     String text = sc.next();
50
51     System.out.print("Pattern: ");
52     String pattern = sc.next();
53
54     List<Integer> matches = findPattern(text, pattern);
55
56     if (!matches.isEmpty()) {
57         for (int idx : matches) {
58             System.out.println("Pattern found at index " + idx);
59         }
60     } else {
61         System.out.println("No match found");
62     }
63
64     sc.close();
65 }
66 }
```

Log Text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6

== Code Execution Successful ==

